CHE 565 -- Homework 5

Soki Sem

April 24, 2026

Problem 1:

Closed-loop response to step disturbance

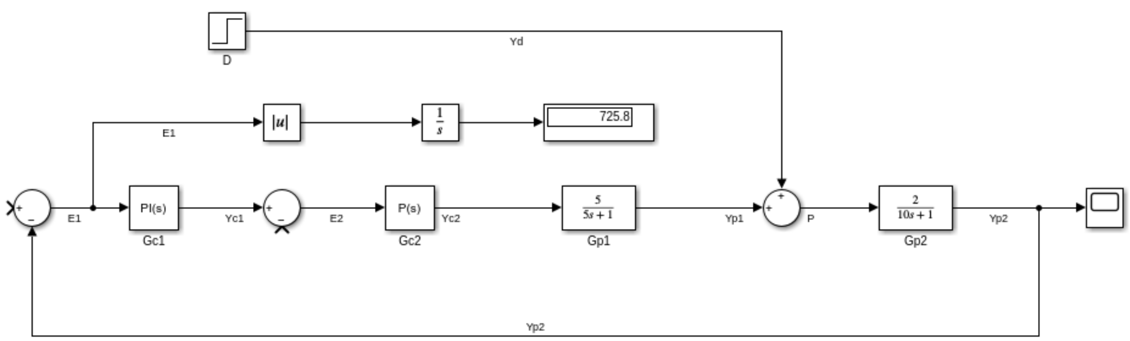

Figure 1: Block diagram of the closed-loop system without cascade control and setpoint and a unit step disturbance from simulink

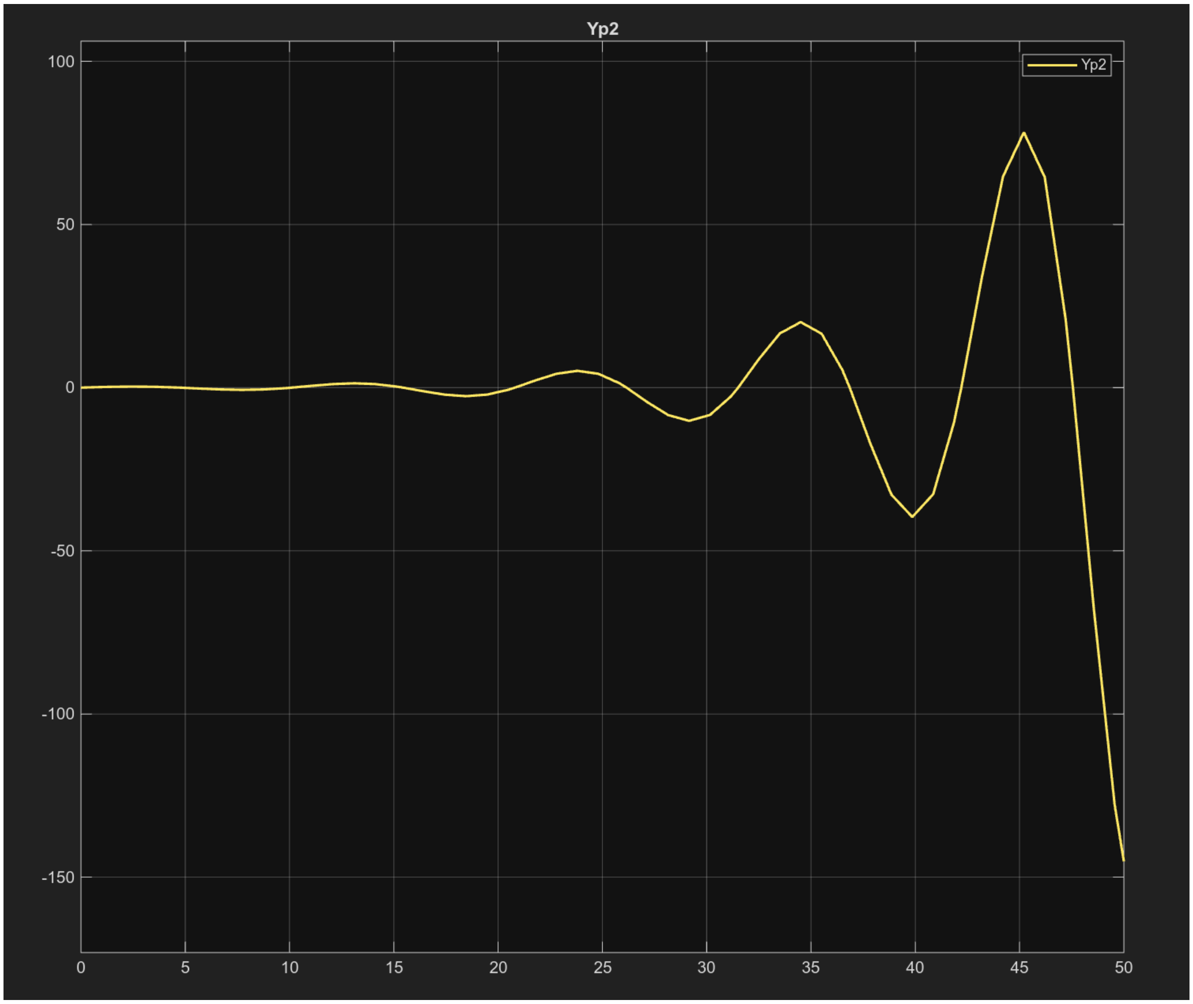

Figure 2: Closed-loop response to no step change in setpoint and a unit step change in disturbance from simulink

Figure 3: Closed-loop response to no step change in setpoint and a unit step change in disturbance, from python

Case 1: Step disturbance, no setpoint change, IAE = 725.3696 (python), IAE = 725.8109 (Simulink)

The PI controller was tuned using the autotuning feature in Simulink P = 0.1837, I = 0.0816 (Simulink autotuning)

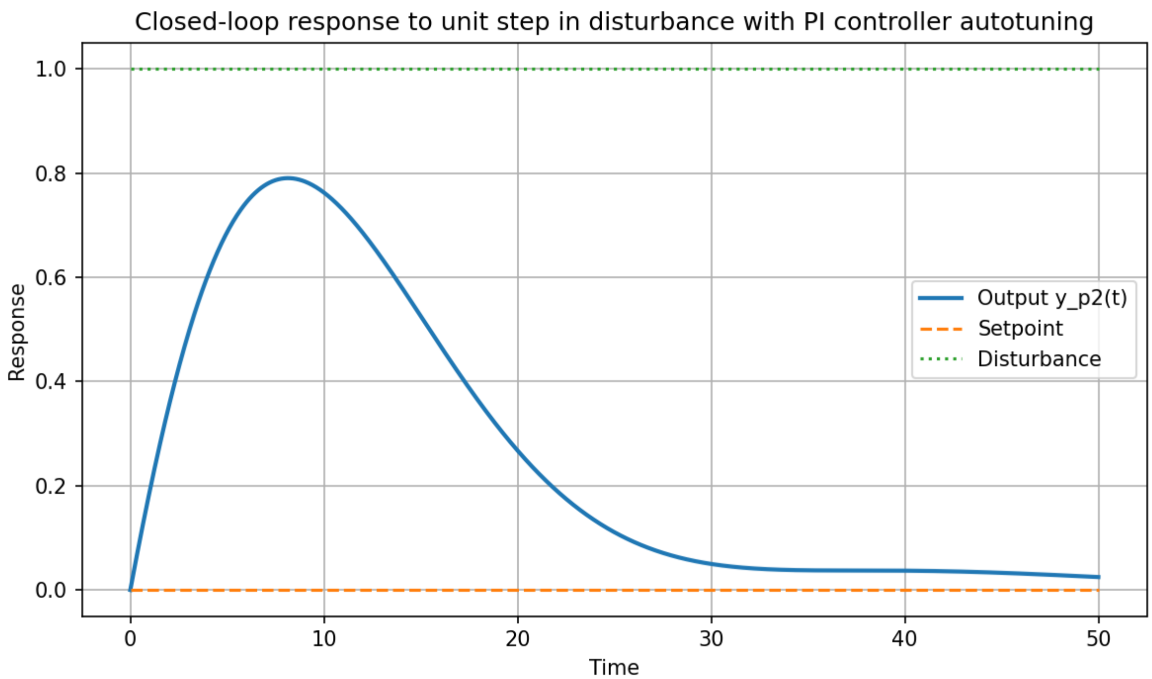

Figure 4: Closed-loop response to no step change in setpoint and a unit step change in disturbance with PI controller autotuning from Simulink

Case 2: Step disturbance, no setpoint change, with PI controller autotuning, IAE = 13.0462 (python), IAE = 13.0463 (Simulink)

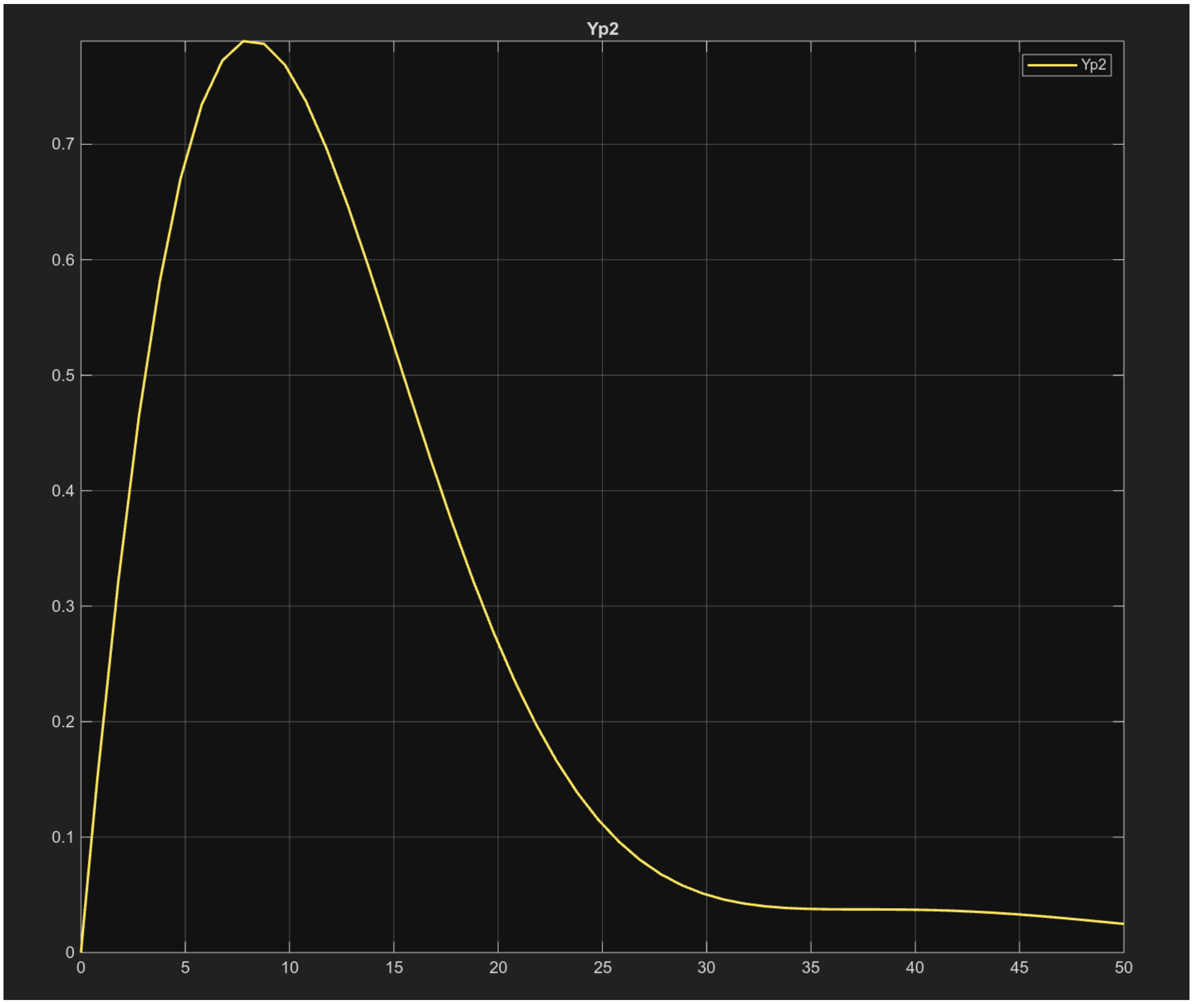

Figure 5: Closed-loop response to no step change in setpoint and a unit step change in disturbance using gains from Simulink autotuning